

ENSEMBLE-BASED PERCEPTUAL AUDIO QUALITY ASSESSMENT WITH CONFIDENCE INTERVALS

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ABSTRACT

Objective audio quality metrics typically provide point estimates, whereas listening tests yield score distributions from which mean opinion scores (MOS), confidence intervals (CIs), and significance decisions are derived. We propose a lightweight intrusive metric that combines a PEAQ-style perceptual front-end (ITU-R BS.1387) with a bagging ensemble of regressors. Calibration with subjective data aligns ensemble outputs with listener scores. The resulting item-dependent score distributions enable uncertainty assessment, panel-size-matched CIs, and identification of less conclusive predictions. Calibration improves agreement with subjective distributions and CI coverage across all evaluated datasets while preserving MOS accuracy. Using only 11 fixed PEAQ features, low-capacity regressors, and public training data, the method performs comparably to more data-intensive end-to-end approaches. Its output can support uncertainty-aware assessment and target listening tests towards uncertain conditions. The distributions also enable approximate pairwise comparisons, but not yet reliable significance inference.

Index Terms— audio quality metrics, uncertainty estimation, ensemble learning, score distributions, listening tests.

1. INTRODUCTION

Perceptual objective audio quality measurement (OAQM) systems are used to benchmark codecs and enhancement algorithms, to monitor quality in communication and streaming [1], and as loss functions [2] or evaluation metrics in machine-learning-based generative audio [3]. Classical OAQMs follow a two-stage design: a signal-processing front-end implements a psychoacoustic model and extracts features describing different degradation dimensions, and a compact machine-learning (ML) stage, i.e., a regressor, maps these features to an overall quality score using subjective mean opinion score (MOS) ratings, or similar quantities, as training targets [4–7].

Subjective listening tests remain the gold standard for quality assessment. In standardised methods [8–10], results are commonly reported as MOS of Basic Audio Quality (BAQ) together with confidence intervals (CIs), reflecting uncertainty of the estimated mean, and with statistical comparisons between conditions. In contrast, most OAQMs perform only point estimation. However, it would be desirable to know the degree of uncertainty associated with a OAQM MOS prediction, whether the corresponding interval achieves adequate empirical coverage (i.e., the estimates’ uncertainty matches the uncertainty achieved by subjective assessment methods), and

whether an estimated quality difference would lead to the same decision as a listening test.

Beyond predicting a single MOS, the primary objective considered in this paper is to estimate the conditional distribution of subjective listener scores for a given reference–signal under test (SUT) pair. Such a distribution would allow a model to simulate listening-test panels, estimate MOS uncertainty, and assess the likelihood of significant differences between conditions. Listener-dependent models such as MBNet [11] and LDNet [12] use individual listener scores to model aspects of listener behaviour, but do not explicitly provide score distribution estimates. In contrast, the Generative Machine Listener (GML) [13] explicitly models a conditional score density, typically by predicting the parameters of a Gaussian or logistic distribution, and samples simulated listener scores from this parametric density. These methods generally rely on deep end-to-end or self-supervised architectures and substantial training resources, while their representations are difficult to relate to established psychoacoustic models.

We propose a non-parametric alternative for estimating subjective score distributions based on an ensemble of bootstrap-trained regressors [14, 15] ¹. For each SUT, the ensemble outputs form a Monte Carlo (MC) sample distribution of MOS predictions rather than the parameters of a prespecified density. This approach is motivated by the assumption that shuffling the MOS-labelled training material produces different plausible learned quality mappings, analogous to listeners with different experience or preferences. We refer to this assumption as the artificial-listener hypothesis. We test whether a simple calibration of the ensemble output based on subjective data is sufficient for ensemble scores to act as proxies for human listener scores. We evaluate this using distributional measures and listening-test-style decisions.

2. METHOD

Fig. 1 illustrates the proposed approach. For each reference signal and SUT comparison, distortion metrics are computed relative to its reference and mapped to MOS by a regressor ensemble.

2.1. System Overview and Perceptual Front-End

Let $\mathbf{x}^{\text{ref}}, \mathbf{x}^{\text{sut}} \in \mathbb{R}^{T \times C}$ denote the reference and SUT digital audio signals with T time samples and C channels. A non-trainable intrusive PEAQ-style front-end² extracts a feature vector $\mathbf{z} = F_{\text{PEAQ}}(\mathbf{x}^{\text{ref}}, \mathbf{x}^{\text{sut}}) \in \mathbb{R}^J$ shared by all regressors. We use

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¹Companion page is provided at: <http://audiolabs-erlangen.de/resources/2027-ICASSP-ELQA> (in preparation after review)

²The PEAQ front-end implementation is available at: <https://www.mmsp.ece.mcgill.ca/Documents/Software/>.

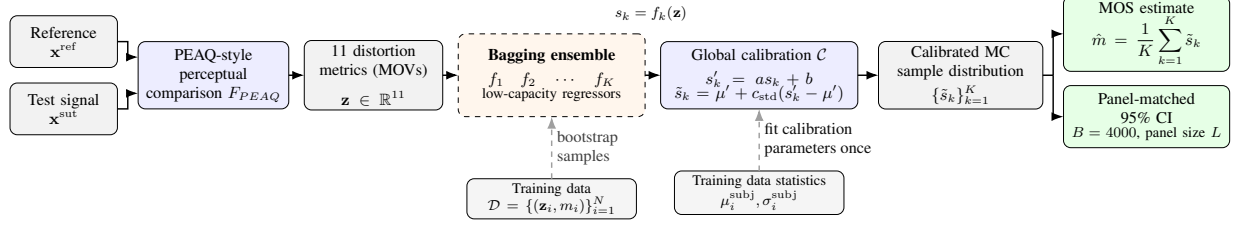


Fig. 1: Proposed system. A shared PEAQ-style front-end feeds bootstrap-trained regressors. Affine mean and global width calibration convert their outputs into candidate artificial-listener scores, from which MOS and panel-matched CIs are estimated. Solid and dashed arrows denote inference and training or calibration, respectively.

the $J = 11$ model output variables (MOVs) of the basic PEAQ model [4, 16]. They describe perceptually relevant degradations, including bandwidth limitation, noise, and temporal or spectral distortions, and are computed once for each signal pair.

The feature vector is passed to K low-capacity regressors, producing $s_k = f_k(\mathbf{z}) \in \mathbb{R}$ for $k = 1, \dots, K$. The regressors may use one or more multivariate regression algorithms and are trained on different bootstrap samples. Before calibration, the variation among their outputs reflects the sensitivity of the learned MOS mapping to the bootstrap-resampled training data. After calibration, the resulting score distribution is expected to more closely resemble subjective score distributions. A global calibration $\mathcal{C} : \mathbb{R}^K \rightarrow \mathbb{R}^K$ converts the raw member scores into calibrated scores $\tilde{s}_k$:

$$(\mathbf{x}^{\text{ref}}, \mathbf{x}^{\text{sut}}) \xrightarrow{F_{PEAQ}} \mathbf{z} \xrightarrow{\{f_k\}_{k=1}^K} \{s_k\}_{k=1}^K \xrightarrow{\mathcal{C}} \{\tilde{s}_k\}_{k=1}^K. \quad (1)$$

2.2. Ensemble Training and Monte Carlo Resampling

For each training stimulus $i = 1, \dots, N$, the front-end produces a feature vector $\mathbf{z}_i$. Given its corresponding L_i listener scores $y_{i\ell} \in [0, 100]$ ³, $\ell \in \{1, \dots, L_i\}$, the subjective MOS representing the regression target is $m_i = L_i^{-1} \sum_{\ell=1}^{L_i} y_{i\ell}$. Here, m_i denotes the observed subjective MOS for stimulus i , viewed as an estimate of the corresponding population mean. The regression training data are therefore $\mathcal{D} = \{(\mathbf{z}_i, m_i)\}_{i=1}^N$.

For each ensemble member k , we draw a bootstrap sample (MC resample) $\mathcal{D}^{(k)}$ of size N from $\mathcal{D}$ with replacement and fit each regressor by squared-error minimisation.

For each stimulus, the mean μ^{mc} and standard deviation σ^{mc} of the ensemble output scores could be used as a point estimate of subjective MOS and a measure of score variability induced by the training-data partitions, respectively [17]. However, this distribution does not necessarily reflect subjective-score distributions accurately. Raw ensemble means can be biased or incorrectly scaled relative to subjective scores, and raw ensemble widths need not match subjective score variability. We therefore estimate one global mean-and-width calibration using the training database as ground truth to be applied to the raw score distribution.

2.3. Global Mean-and-Width Calibration

In order to align location and scale parameters of the MC distributions to the distribution of subjective scores, we first apply an affine transformation with global parameters a and b to each ensemble-member score: $s'_k = as_k + b$. This transformation maps the ensemble mean according to $\mu' = a\mu + b$ and scales the standard deviation

of the ensemble-member scores by $|a|$. The parameters are fitted by least squares:

$$(a, b) = \arg \min_{a, b} \sum_i [m_i - (a\mu_i^{\text{mc}} + b)]^2. \quad (2)$$

To align the scale parameters of the ensemble-score distribution with the observed listener-score spread, the deviations of the affinely mapped member scores are rescaled

$$\tilde{s}_k = \mu' + c_{\text{std}} (s'_k - \mu'), \quad k = 1, \dots, K, \quad (3)$$

where $\mu' = K^{-1} \sum_k s'_k$. This operation preserves the affine-calibrated arithmetic mean μ' while scaling the standard deviation of the member scores by c_{std} . We estimate this global width factor as

$$c_{\text{std}} = \text{median}_i \frac{\sigma_i^{\text{subj}}}{|a| \sigma_i^{\text{mc}}}, \quad (4)$$

where σ_i^{subj} is the sample standard deviation of the individual listener scores for stimulus i . We assume that both subjective and ensemble variances are finite and strictly positive, as is the case for the data used in the experiments.

In the proposed two-stage transformation Eq. (2) aligns the predicted mean, while Eq. (4) sets one global spread factor such that the median ratio of subjective to calibrated ensemble standard deviation across the calibration stimuli equals one. From the model outputs $\tilde{s}_k$ forming an estimated empirical score distribution, point estimates $\hat{m} = \frac{1}{K} \sum_{k=1}^K \tilde{s}_k$ and confidence intervals can be calculated.

2.4. Panel-Size-Matched CI Estimation

To compare the calibrated model distribution with a listening test containing L listeners, we simulate $B = 4000$ pseudo-panels of the same size. For the b th bootstrap draw, calibrated member scores are sampled with replacement and averaged:

$$\bar{s}^{(b)}(L) = \frac{1}{L} \sum_{\ell=1}^L \tilde{s}_{I_{b\ell}}, \quad I_{b\ell} \sim \mathcal{U}\{1, \dots, K\}. \quad (5)$$

Here, $\mathcal{U}\{1, \dots, K\}$ denotes the discrete uniform distribution over the K ensemble-member indices. The nominal 95% panel-matched ensemble CI is given by the empirical quantiles of the resulting pseudo-panel means⁴:

$$\text{CI}_{i,95}^{\text{mc}}(L) = [Q_{\tilde{s}_i(L)}(\alpha), Q_{\tilde{s}_i(L)}(1 - \alpha)] \quad (6)$$

with $\alpha = 0.025$. Under the artificial-listener hypothesis, this interval estimates the variation that would be obtained by repeatedly drawing panels of L calibrated model listeners.

³This particular scale range corresponds to the subjective listening test method of choice for the ground-truth data.

⁴ $Q_{\tilde{s}_i(L)}(p)$ denotes the empirical p -quantile of the Monte Carlo distribution of panel-mean scores $\bar{s}_i(L)$ for stimulus i and panel size L .

3. EXPERIMENTS

To assess the artificial-listener hypothesis, we evaluate three increasingly demanding implications: empirical MOS coverage, agreement of marginal score distributions, and agreement of paired significance decisions.

3.1. Training and Evaluation Databases

The ensemble and global calibration are trained on SEBASS-DB [18], a public meta-dataset of MUSHRA [10] listening tests on the BAQ of blindly separated audio sources. SEBASS-DB combines SASSEC, SiSEC08, SAOC-DB, and SiSEC18, providing more than 11 000 ratings for over 900 separated signals. It was selected for its public availability.

For cross-domain evaluation in the audio-coding domain, we use the Open Dataset of Audio Quality (ODAQ) [19] and the Unified Speech and Audio Codec (USAC) verification sets [20]. Preliminary training on ODAQ was less effective, possibly because its emphasis on isolated coding artefacts provides fewer examples of the artefact interactions encountered in practice and needed by the regressors. Both evaluation sets contain stereo reference excerpts comprising music, speech, dialogue, and movie-like soundtracks, processed using different degradations (*conditions*). All datasets provide reference signals, degraded signals, and listener scores on a 0 (*bad*) to 100 (*excellent*) quality scale.

3.2. Base Learner Selection

Multivariate Adaptive Regression Splines (MARS) were selected as the base learner for their accuracy–complexity trade-off and smooth behaviour [17]. Learner selection and hyperparameter tuning used repeated 50/50 holdout resampling on SEBASS-DB; the final ensemble members were fitted on bootstrap resamples as described in Sec. 2.2. For these experiments, $K = 50$ base regressors provided a practical accuracy–complexity trade-off, yielding sufficiently stable predictions without unnecessary computational cost. Experiments used Python 3.10, `scikit-learn` 1.7.2, and `py-earth2` 0.2.0.⁵

3.3. Operational Evaluation

All analyses are performed separately by database or listening-test experiment for raw (R) and calibrated (C) outputs.

Distributional Agreement: As a compact measure of listener-score distribution agreement, we compute the two-sample Kolmogorov–Smirnov (KS) statistic separately for every stimulus i . The KS statistic is defined as

$$D_{KS,i} = \sup_x \left| F_i^{\text{subj}}(x) - F_i^{\text{mc}}(x) \right|, \quad (7)$$

where F_i^{subj} and F_i^{mc} are the empirical cumulative distribution functions (CDFs) of the subjective and ensemble scores for stimulus i . A smaller value indicates less worst-case mismatch between CDFs and reduces the risk of locally unreliable proxies.

For each database, we report the mean per-stimulus KS statistic, $\bar{D}_{KS} = \frac{1}{N} \sum_{i=1}^N D_{KS,i}$.

MOS Accuracy and Empirical CI Coverage: For conventional OQM evaluation, we report the point-estimate MOS comparison between subjective and objective means [22]. Model outputs are clipped to this range, except in the significant-difference

analysis, where clipping is omitted to avoid boundary-induced effects. The Pearson correlation coefficient (PCC) and the root mean squared error (RMSE) are computed from the per-item subjective and ensemble means. We also report DeePAQ and the 2f metric [23] as a comparable state-of-the-art baseline, as 2f’s regressor was likewise fitted on SEBASS-DB. Results for other state-of-the-art metrics on these datasets are provided in [21].

For each item, the objective CI is the panel-matched bootstrap interval in (6), with $L = L_i$. Its empirical coverage is

$$\hat{C}_{95} = \frac{1}{N} \sum_{i=1}^N \mathbb{I} [m_i \in \text{CI}_{i,95}^{\text{mc}}(L_i)]. \quad (8)$$

Thus, $\hat{C}_{95}$ is the fraction of items for which the panel-matched ensemble CI contains the corresponding subjective MOS⁶. Values closer to the nominal level of 0.95 indicate better coverage.

Listening-Test-Style Significant Differences: For each experiment, listener scores are first averaged across items for every listener–condition pair. A one-factor repeated-measures ANOVA with factor *condition* is then computed, followed by paired t -tests for all condition pairs with Bonferroni correction at $\alpha = 0.05$.

The same procedure is applied to the ensemble: scores are averaged across items for every ensemble-member–condition pair. Because the number of ensemble members can differ from the number of listeners, ensemble members are bootstrapped with replacement to the subjective panel size. A pair is regarded as objectively significant when at least half of the bootstrap repetitions yield a Bonferroni-corrected significant difference.

Let $h_p \in \{-1, 0, +1\}$ denote the subjective result for pair p , where 0 denotes a non-significant difference and the sign denotes the preferred condition. Let $\hat{h}_p$ be the corresponding objective decision. We report directional significant-pair recall and false-positive rate,

$$R_{\text{sig}} = \frac{\sum_p \mathbb{I}[h_p \neq 0 \wedge \hat{h}_p = h_p]}{\sum_p \mathbb{I}[h_p \neq 0]}, \quad (9)$$

$$\text{FPR} = \frac{\sum_p \mathbb{I}[h_p = 0 \wedge \hat{h}_p \neq 0]}{\sum_p \mathbb{I}[h_p = 0]}. \quad (10)$$

High R_{sig} indicates that subjectively significant differences are detected with the correct direction, whereas low FPR indicates that the system rarely reports a difference for a subjectively non-significant pair. These metrics must be interpreted jointly: reporting more pairs as significantly different may increase recall but can also increase the false-positive rate.

4. RESULTS AND DISCUSSION

4.1. Accuracy, Coverage, and Distributional Agreement

Tab. 1 reports point estimate MOS accuracy, empirical CI coverage, distributional agreement, and significance-decision agreement. Calibration improves coverage on every dataset: from 0.28–0.31 to 0.69–0.82 on the SEBASS-DB subsets, from 0.34 to 0.47 on ODAQ, and from 0.14–0.22 to 0.33–0.43 on USAC. It also reduces $\bar{D}_{KS}$ throughout, indicating better agreement between the predicted and subjective score distributions (see, e.g., Fig. 2). Coverage nevertheless remains below the nominal 0.95, particularly on USAC. This may partly reflect our deliberate use of the publicly available

⁵<https://scikit-learn.org/>; <https://pypi.org/project/py-earth2/>.

⁶The indicator function $\mathbb{I}[\cdot]$ equals 1 if its argument is true and 0 otherwise.

Table 1: Results for MOS accuracy, uncertainty, and decisions. R/C: raw/calibrated ensemble. PCC entries are R/C/2f/DeePAQ; RMSE entries are R/C/2f; all others are R/C. DeePAQ PCC values are from [21]. Pair counts are all/significant; ODAQ pools six experiments. Coverage is below 0.95, so higher is better. Bold marks the best value per dataset and metric.

Dataset	Pairs all/sig.	Scalar MOS Accuracy (R/C/2f/DeePAQ)		Uncertainty (R/C)		Decision Agreement (R/C)	
		PCC $\uparrow$	RMSE $\downarrow$	$\hat{C}_{95} \uparrow$	$\bar{D}_{KS} \downarrow$	$R_{sig} \uparrow$	FPR $\downarrow$
SAOC-DB	153/16	0.93/0.93/0.81/0.93	5.2/ 5.2 /8.1	0.30/ 0.82	0.50/ 0.33	0.94 /0.69	0.84/ 0.60
SASSEC	55/40	0.97/0.97/0.90/0.92	4.4/ 4.4 /7.4	0.31/ 0.75	0.46/ 0.28	1.0 /0.85	0.80/ 0.27
SiSEC08	36/27	0.96/0.96/0.91/0.95	4.6/ 4.6 /6.7	0.28/ 0.69	0.47/ 0.30	1.0 /0.89	0.78/ 0.22
ODAQ	87/82	0.85/0.83/0.87/ 0.92	11/12/ 10	0.34/ 0.47	0.44/ 0.41	0.57 /0.29	0.40/ 0.20
USAC1	36/34	0.94/0.93/0.86/0.90	5.2 /5.7/7.9	0.14/ 0.33	0.43/ 0.36	1.0 /0.94	0.50/0.50
USAC2	21/20	0.81/0.80/0.64/ 0.88	8.1 /8.4/11	0.20/ 0.43	0.41/ 0.34	0.90 /0.85	1.0/ 0.00
USAC3	28/27	0.78/0.77/0.82/ 0.93	9.6/9.5/ 8.5	0.22/ 0.34	0.42/ 0.35	0.74 /0.70	1.0/ 0.00

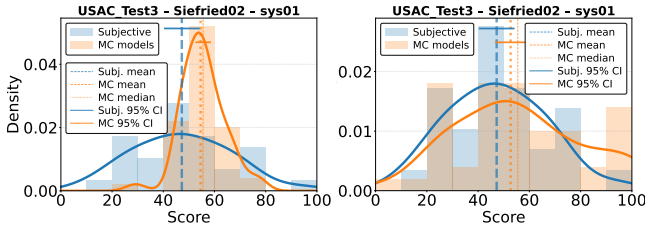


Fig. 2: Example outputs of ensemble score distributions and corresponding subjective MOS distribution for one item in the evaluation database. Left: before calibration ($D_{KS} = 0.48$) and right: after affine mean alignment and width adjustment ($D_{KS} = 0.21$).

SEBASS-DB for reproducible training, at the expense of a mismatch with the codec, bitrate, and content conditions represented in USAC.

Affine calibration has little effect on point estimate MOS accuracy; the small PCC and RMSE changes result from clipping member scores to the MUSHRA scale. Compared with 2f, the ensemble mean achieves higher PCC and lower RMSE on every non-aggregated dataset except USAC3, while 2f has a modest advantage on pooled ODAQ. Unlike the point estimate-only 2f model, the ensemble also provides item-specific score distributions and panel-size-matched CIs. On the common USAC1–3 sets, its calibrated PCC/RMSE values are 0.93/5.67, 0.80/8.39, and 0.77/9.52, whereas GML with CutMix reports PCCs of 0.88, 0.89, and 0.92, respectively [13]. Thus, although our method is competitive on USAC1, it is weaker than GML on USAC2 and USAC3. Nevertheless, these results are promising given that our lightweight approach uses only 11 fixed PEAQ features and low-capacity MARS regressors trained on the public SEBASS-DB (11 000 ratings), whereas GML uses 67 505 individual ratings from an internal, codec-oriented corpus and a 15.25-million-parameter network trained on an NVIDIA A100 GPU. The performance achieved despite the mismatch between SEBASS-DB and the codec, bitrate, and content conditions in USAC suggests that training on sufficiently representative public data could further improve generalization.

4.2. Pairwise Decision Agreement

Calibration makes the paired tests more conservative: FPR decreases or remains unchanged, but significant-pair recall decreases on every dataset. These rates can be unstable when based on few pairs: SAOC-DB has only 16 subjectively significant pairs, while USAC1–

3 have only two, one, and one non-significant pairs, respectively.

This trade-off follows from the width adjustment. For conditions A and B , calibration preserves their mean contrast while scaling the centered member variation in paired differences by c_{std} . For the complete, unresampled ensemble and without clipping, $t_{cal} = t_{aff}/c_{std}$, where t_{aff} denotes the statistic after affine mean calibration. Because the fitted $c_{std} > 1$, significance becomes less likely. Matching marginal score distributions is therefore insufficient for paired decisions, which also depend on cross-condition covariance.

Reliable listening-test inference would require separate calibration of marginal variability and cross-condition covariance using listener-level data. Artificial listener panels should retain the current panel-size matching while also preserving within-listener dependencies and the target test design. Calibration and decision thresholds should then be validated on independent listening tests for CI coverage, false-positive control, and statistical power.

Although the ensemble cannot yet replace panel-level hypothesis testing, it provides an uncertainty-aware OAQM: its mean estimates MOS, while its item-dependent spread and approximate CIs help identify predictions requiring subjective validation.

5. SUMMARY AND CONCLUSIONS

We presented a lightweight, distribution-aware OAQM that produces empirical score distributions and panel-size-matched CIs in addition to MOS estimates. Mean-and-width calibration improves CI coverage and distributional agreement on every evaluated dataset while maintaining competitive MOS accuracy. Unlike point-estimate metrics, the method provides item-dependent uncertainty; unlike deep generative listener models, it relies on only 11 PEAQ features, low-capacity regressors, and exclusively public training data. We therefore consider ensemble regressors a practical and valuable approach for uncertainty-aware quality assessment, with clear potential for improvement through more diverse training data and refined calibration. However, calibrated marginal distributions do not ensure reliable paired significance decisions because cross-condition dependencies are not calibrated. The system is therefore useful for uncertainty screening and prioritising listening tests, but cannot yet replace panel-level inference. Future work will investigate listener-level calibration of cross-condition covariance and condition-dependent uncertainty using independent listening tests. Incorporating out-of-domain detection may further enable the system to defer automatically when its uncertainty estimates are unlikely to transfer.

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